



DILEEP KUMAR THIRUVENKADAM

Heilbronn, Germany | +49 15560729644 | dileepkumarthiruvenkadam@gmail.com

[LinkedIn](#) | dileepkumarthiruvenkadam.de

SUMMARY

Software Engineering Master's student at Hochschule Heilbronn with a focus on DevSecOps, cloud infrastructure, and automation. With professional experience in backend development and test orchestration, I bridge the gap between development and operations. My academic performance includes a 1.0 grade in Cloud Computing and DevOps & SecOps. I am proficient in Python, Java, and JavaScript frameworks.

EDUCATION

Hochschule Heilbronn, Heilbronn, BW, Germany 09/2024-Present

- Current Average Grade: 2.2
- Notable Achievements: Grade 1.0 in Cloud Computing and DevOps & SecOps; Grade 1.3 in Advanced Software Architecture and Data Science Lab

SRM Institute of Science and Technology, Chennai, TN, Germany. 06/2018-05/2022

Final Grade: 1.9 (German Scale)

WORK EXPERIENCE

Quaking Aspen Private Limited, Associate Software Developer, Chennai, TN, India. 08/2022-07/2024

- Developed and optimized core backend modules on AWS with Java and Python, resulting in a 15% reduction in system latency through improved query logic
- Engineered the architecture of Selenium and Cypress regression suites that cut manual test efforts by 40%, enabling consistent release delivery across a 32-sprint, 96-week programme
- Built and documented RESTful APIs connecting front-end services to PostgreSQL databases, standardizing integration contracts across 5 unique API endpoints and reducing cross-team onboarding friction. Facilitated team discussions to identify and resolve key challenges, leading to improvement in performance
- Led sprint planning and retrospective facilitation within an Agile Scrum team; owned task lifecycle tracking in Jira and enforced Git branching standards to maintain codebase stability
- Streamlined CI/CD pipelines on AWS, cutting average build-to-deploy cycle time by 40% and supporting 18 production deployments per sprint
- Reviewed 500+ pull requests across the engineering team, enforcing coding standards that reduced post-merge defects by 30%

Zoho Corporation, Web Development Intern, Chennai, TN, India 01/2021-06/2021

- Designed and implemented responsive UI components using ReactJS, focusing on efficient state management and API data rendering
- Performed unit and integration testing to ensure cross-browser compatibility and UI consistency
- Delivered 12 production-ready UI features within a 5-member front-end team, meeting sprint deadlines across a 6 month engagement

PROJECTS & LEADERSHIP EXPERIENCE

COVID-19 Detection from Chest X-rays, Bachelor's Final year project 12/2021-05/2022

- Developed a Convolutional Neural Network (CNN) to classify lung X-ray images into COVID-positive, pneumonia, and normal cases, preprocessing datasets with OpenCV and image augmentation.
- Evaluated and visualized performance metrics—including accuracy, precision, recall, F1 score, and confusion matrices—using Matplotlib and Seaborn to ensure model reliability.

Automation for Workflow Optimization, Quacking Aspen, Chennai, TN 08/2022-02/2023

- Developed Python automation scripts to reduce manual tasks and improve operational efficiency. Technologies: Python, shell scripting, and cron jobs.

CI/CD Pipeline Integration, Hochschule Heilbronn (HHN) - Devops, Heilbronn, BW, Germany 09/2024-01/2025

- Integrated GitHub Actions for automated testing, deployment, and version control. Technologies: GitHub Actions, Docker, Terraform, AWS

Cloud Migration & Monitoring, Hochschule Heilbronn (HHN) - Devops, Heilbronn, BW, Germany 03/2024-07/2025

- Migrating services to AWS and setting up monitoring/logging tools. Technologies: AWS CloudWatch, Terraform, Docker

Intelligent Traffic Simulation System, HOCHSCHULE HEILBRONN - Realtime 3D Engine, Heilbronn, BW, Germany 09/2025-02/2025

- Developed a traffic flow modeling tool in Python to simulate real-world congestion scenarios and optimize urban routing algorithms.

SKILLS

Languages: English (Fluent), German (Intermediate), Tamil (Native)

Skills: Vulnerability Scanning, Testing, Terraform, SQL (PostgreSQL, MySQL), SonarQube (Static Analysis), Snowflake, Shift-Left Security, Selenium, Scikit-learn, RestAssured, Regression, Python, PyTest, Power BI, Pandas, NumPy, Manual UAT, Kubernetes, JUnit, Jira, JavaScript (Node.js, Express, ReactJS), Java (Spring Boot), Docker, Databricks, Cypress, CI/CD Pipeline Design, Bash scripting, and AWS (S3, CloudWatch, SNS, IAM, Elastic Bean)